EEE 554 – Probability and Stochastic Processes

Homework Assignment #3

Assigned: 5 March 2026
Due: 2 April 2026

Reading (from Papoulis and Pillai):

Chapter 7, pp. 249–250; Chapter 8.

Work to hand in:

1. A complex $n \times n$ matrix $A = [a_{ij}]$ is positive definite.
   (a) Show that elements a_{ii} on the main diagonal of A are positive.
   (b) Show that the eigenvalues of $G = A^\dagger A$ and $F = AA^\dagger$ are the same. If $\{E_j\}$ are the (right) eigenvectors of G, what are the eigenvectors of F?

2. Recall that a complex $n \times n$ matrix U is *unitary* if $U^\dagger U = \mathbb{I}$ where $U^\dagger$ denotes the hermitian transpose of U and $\mathbb{I}$ is the $n \times n$ identity matrix.
   (a) Verify that $\mathbb{I}$ is unitary.
   (b) If U is unitary, show that U is invertible and that U^{-1} is also unitary.
   (c) If U and V are both unitary, show that UV is unitary.
   (d) When U and V are unitary with $\det(U) = \det(V) = 1$, what is $\det(UV)$?

   The first three of these properties show that the collection of unitary matrices is a *group*, usually denoted $\mathbf{U}(n)$. The fourth property shows that the collection of all unitary matrices with determinant one is also a group, a subgroup of $\mathbf{U}(n)$, called the *special unitary group* and denoted $\mathbf{SU}(n)$.

3. If M is a $n \times n$ positive definite hermitian matrix, there is a unique $n \times n$ positive definite hermitian matrix $M^{1/2}$ such that $M^{1/2}M^{1/2} = M$. $M^{1/2}$ is the *square root* of M, which is computed by the Matlab `sqrtm` function. Denote by $M^{-1/2}$ the square root of M^{-1}.
   (a) If a complex random vector $\mathbf{X}$ has positive definite covariance matrix $C_{\mathbf{X}}$ and $V = C_{\mathbf{X}}^{-1/2}$, find the covariance matrix $C_{\mathbf{Y}}$ of $\mathbf{Y} = V\mathbf{X}$.
   (b) If $\mathbf{X}$ has covariance matrix $\mathbb{I}$ and $V = C^{1/2}$ where C is an arbitrary $n \times n$ positive definite hermitian matrix, find the covariance matrix $C_{\mathbf{Y}}$ of $\mathbf{Y} = V\mathbf{X}$.

4. Let $A = [a_1 \ \dots \ a_n]$ and consider a random vector $\mathbf{X} = [\mathbf{x}_1 \ \dots \ \mathbf{x}_n]^\mathsf{T}$ with the RVs $\mathbf{x}_k$ iid.
   (a) Under what condition on A will $\hat{\boldsymbol{\mu}} = A\mathbf{X}$ be an unbiased estimator of $\mu = \mathsf{E}[\mathbf{x}]$?
   (b) Calculate $\text{var}(\hat{\boldsymbol{\mu}})$.
   (c) Determine A so that $\hat{\boldsymbol{\mu}}$ is unbiased and has minimal variance. Find the value of this minimum variance. [Hint: Do the $n = 2$ case first.]

5. Problem 8-14, page 368.

6. Problem 8-20, page 368.

7. Problem 8-21, page 368.

8. Problem 8-24, page 368.

9. (Computer problem) Suppose $\mathbf{Y} \sim \mathcal{N}[\mathbf{0}, C_{\mathbf{Y}}]$ with

$$C_{\mathbf{Y}} = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

   (a) Generate $M = 10,000$ independently realized Gaussian 3-vectors $\mathbf{X}_1, \ldots, \mathbf{X}_M$ drawn from a $\mathcal{N}[\mathbf{0}, \mathbb{I}]$ distribution and verify that these data have (approximately) the desired mean and covariance.

   (b) Find a 3×3 matrix V so that $\mathbf{Y} = V\mathbf{X}$ has the distribution specified above.

   (c) Transform the data generated in part (a) to obtain M samples of $\mathbf{Y}$. Verify that these data have (approximately) the desired mean and covariance.

10. (Computer problem) The data for this problem are in the file bpsk.dat, which can be downloaded from Blackboard. A binary communication signal (bit stream) is corrupted by the addition of white Gaussian noise having mean $\mu = 3$ and variance $\sigma^2 = 0.05$; i.e., each datum is of the form $\mathbf{x}_k = s_k + \mathbf{n}_k$ where $s_k = 1$ with probability $P(H_1)$, $s_k = 0$ with probability $P(H_0)$, and $\mathbf{n}_k \sim \mathcal{N}[3, 0.05]$.

   (a) Assume $P(H_1) = 0.48$ and find the detection rule yielding the minimum probability of error. What is the minimum achievable bit error rate?

   (b) Apply your detection rule to the data provided and convert the resulting bit stream to ASCII text (Matlab m-files bits2text and text2bits are provided to help with the conversions here and in part (c)). How many bit errors do there seem to be in the decoded message? Is this consistent with your theoretical prediction?

   (c) Add just enough zero-mean white Gaussian noise to the data to increase the bit error rate to 0.125. Verify that the desired error rate is (approximately) achieved. How does the decoded message look at this SNR?

11. (Computer problem) Data files hw3xr.dat, hw3yr.dat, and hw3zr.dat contain, respectively, the real parts independently realized samples of a complex random vector $\mathbf{Z} = [\mathbf{x}\ \mathbf{y}\ \mathbf{z}]^{\mathsf{T}}$. The files hw3xi.dat, hw3yi.dat, and hw3zi.dat contain the corresponding imaginary parts.

   (a) Estimate the covariance matrix $C_{\mathbf{Z}}$ of $\mathbf{Z}$.

   (b) Find a unitary matrix U so that $\mathbf{W} = U\mathbf{Z}$ is an uncorrelated random vector (i.e., $C_{\mathbf{W}}$ is diagonal).

   (c) Apply U to the data provided to obtain samples of $\mathbf{W}$ and verify that the covariance matrix estimated from these data is (approximately) diagonal.